Skills Summary

Surface Area, Composite Shapes, and Scale Design

Use this reference while completing your worksheet or when studying.

Surface Area of a Rectangular Prism

Formula: $SA = 2(\ell w + \ell h + wh)$ — three pairs of matching faces (top/bottom, front/back, two ends). A cube shortcut: $SA = 6s^2$. For an *open* container, list the faces that exist and leave out the missing one.

Example: Find the surface area of a closed $5 \times 2 \times 3$ cm box.
Step 1. List the three face pairs from length, width, and height, then double their sum.
Step 2. $SA = 2(5 \cdot 2 + 5 \cdot 3 + 2 \cdot 3) = 2(10 + 15 + 6) = 2 \cdot 31 = 62$.
$\Rightarrow$ **62** cm^2

Try it: Find the surface area of a cube with 4-inch edges.

check: **96** in^2

Area of a Composite Shape

Strategy: cut the shape into rectangles (and triangles) along its corners, find each piece's area, and add. Read each piece's side lengths off the corner coordinates by subtracting.

Example: An L-shape has corners $(0,0)$, $(6,0)$, $(6,3)$, $(2,3)$, $(2,5)$, $(0,5)$. Find its area.
Step 1. Cut along $y = 3$: a 6×3 bottom rectangle and a 2×2 top-left piece.
Step 2. $6 \cdot 3 = 18$ and $2 \cdot 2 = 4$; total $18 + 4 = 22$.
$\Rightarrow$ **22** square units

Try it: Find the area of the L-shape with corners $(0,0)$, $(7,0)$, $(7,2)$, $(3,2)$, $(3,6)$, $(0,6)$.

check: **26** square units

Scale Drawings: Lengths

Rule: a scale like 1 cm : 6 m tells you what one drawing unit is worth. Drawing $\rightarrow$ real: *multiply* by the scale. Real $\rightarrow$ drawing: *divide* by the scale.

Example: Scale 1 cm : 6 m. A hallway is 4.5 cm on the drawing. How long is the real hallway?
Step 1. Drawing to real — multiply the drawing length by the scale value.
Step 2. $4.5 \cdot 6 = 27$.
$\Rightarrow$ **27** m

Try it: Scale 1 in : 9 ft. A porch is 2.5 in on the plan. Find the real length.

check: **22.5** ft

Scale Factors and Area

Rule: when every length is multiplied by scale factor k, area is multiplied by k^2 — both the length *and* the width stretch. Doubling lengths quadruples area; a $1:20$ model has $\frac{1}{400}$ the floor area.

Example: A sticker of area 5 cm^2 is enlarged by scale factor 4. Find the new area.
Step 1. Area scales by the *square* of the factor, so multiply by 4^2, never by 4.
Step 2. $5 \cdot 4^2 = 5 \cdot 16 = 80$.
$\Rightarrow$ **80** cm^2

Try it: A photo of area 2.5 in^2 is enlarged by scale factor 10. Find the new area.

check: **250** in^2